⑧ Dependability modeling via Markov Chains

Intro: For Analyzing Dependability

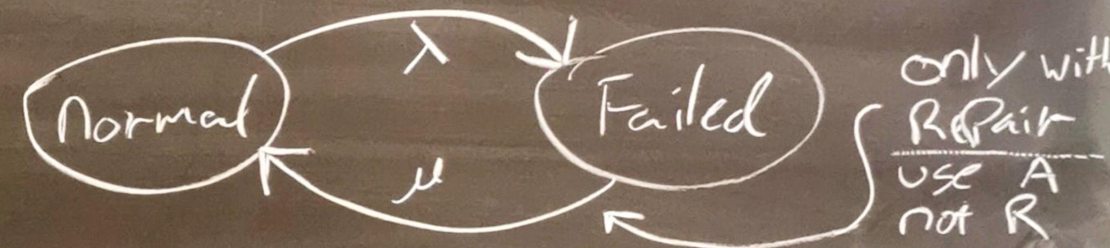
- Availability: Readiness for Correct service

  $A(t) := P_r(\text{sys in normal state at time } t)$

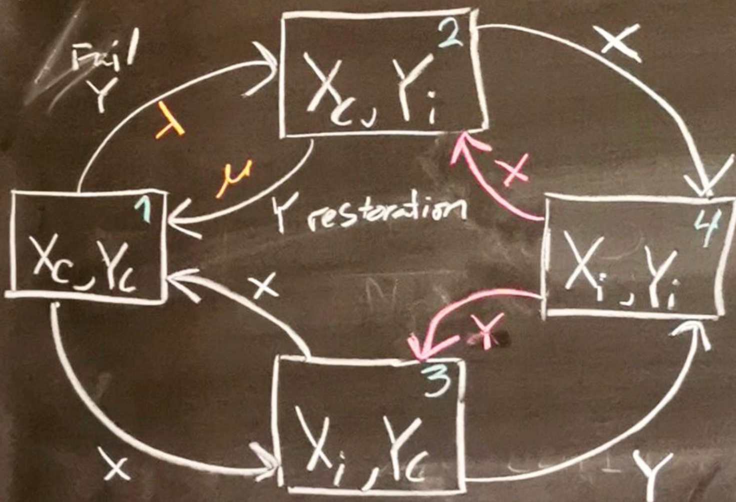
- Reliability: Continuity of correct service

  $R(t) := P_r(\text{sys in normal state in time } [0,t])$

Binary model ⟶



2 Component Product State Space examples: Redundant



Correct: 1 2 3 failed: 4

c = correct i = incorrect

A – with RED

R – without RED

$$\underline{Q} = \begin{bmatrix} -(\lambda_x+\lambda_y) & \lambda_y & \lambda_x & 0 \\ \mu_Y & -(\lambda_x+\mu_Y) & 0 & \lambda_x \\ \mu_x & 0 & -(\lambda_Y+\mu_x) & \lambda_Y \\ 0 & \mu_X & \mu_Y & -(\mu_x+\mu_y) \end{bmatrix}$$

Prob of being in States at time t

$\underline{P}(t) = \underline{P}(0)\,\mathrm{expm}(\underline{Q}t)$ (Cont.)

$A(t) = P_1(t) + P_2(t) + P_3(t)$

$R(t)$ same but without red in Matrix